Capstone Project Brief – DataCo Global

Project Title

Sales & Customer Intelligence Dashboard for DataCo Global

Business Context

DataCo Global is a multinational company that specializes in B2C product sales across various countries and markets. With a rapidly growing customer base and a wide product portfolio, executives are looking to gain deeper insights into:

- Sales performance over time
- Profit contribution by region and category
- How delivery and customer satisfaction indicators impact business outcomes
- Customer segmentation and regional trends

You've been engaged as a **Junior Data Analyst** to transform the company's raw sales data into a clean analytical model and use it to create a professional two-page dashboard in Power BI.

Your Tasks

Data Modeling (Star Schema Required)

Using the provided flat dataset, you are required to build a proper dimensional model following this structure:

Fact Table: FactSales

- FactSales ID
- Order Id
- Order Date

- Shipment Date
- Customer Id *(FK → DimCustomer)*
- Product Card Id *(FK → DimProduct)*
- Location Key *(FK → DimLocation)*
- Sales and order metrics (e.g., Sales, Quantity, Discount, Profit)
- Shipment metrics (e.g., Shipment Days, Delivery Status, Late Risk)

Dimension Tables

- **DimCustomer**
    - Customer ID, Name, Segment, Address, State, Country
- **DimProduct**
    - Product ID, Name, Price, Status, Category, Department
- **DimLocation**
    - Location Key, Market, Region, City, Country, Latitude, Longitude
- **DimDate**
  Create a custom Date Table from scratch using DAX or Power Query. This table must cover all dates present in the dataset and include standard calendar fields like:
    - Date
    - Year
    - Month
    - Quarter
    - Month Name
    - Week Number
    - Day of Week

Dashboard Design Requirements

Page 1: Executive Summary Dashboard (Home Page)

This should include high-level KPIs and insights. Each KPI must also include its Year-over-Year (YoY) percentage change using DAX formulas.

Required KPIs and YoY Metrics:

- **Total Sales**
    - DAX: `Total Sales = SUM(FactSales[Sales])`
    - YoY: % change compared to same period last year
- **Total Profit**
    - DAX: `Total Profit = SUM(FactSales[Order Profit Per Order])`
    - YoY: Compare with last year's total profit
- **Total Quantity Sold**
    - DAX: `Total Quantity = SUM(FactSales[Order Item Quantity])`
    - YoY: Compare with total quantity sold same period last year
- **Average Delivery Time**
    - DAX: `Avg Delivery Time = AVERAGE(FactSales[Shipment Days (Actual)])`
    - YoY: Compare with last year's average delivery days

Other Visuals to Include:

- Sales by Region or Market (Map or Bar Chart)

- Top Performing Products (by Sales or Profit)
- Customer Segment Performance (Bar or Pie Chart)

Slicers (Filters) to Add:

- Region
- Product Category
- Customer Segment
- Date

Page 2: Deep Dive View

This page should offer detailed views for further analysis:

- Sales trend over time (Line Chart)
- Profit breakdown by Category or Department
- Late delivery risk by Region or Category
- Sales by Segment and Region

Download Dataset

Download the DataCo Global Dataset here

Submit Your Work

Submit your completed Power BI report here

Project Deliverables

Submit a `.pbix` file containing:

- A well-modeled data structure (fact and dimensions)
- A custom-built Date Table created from scratch
- Cleaned and transformed tables
- Two pages of visuals (KPI Dashboard and Deep Dive View)
- DAX calculations for each KPI and corresponding YoY metrics
- Clean, professional, and user-friendly layout